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Interactive report

Optical recording of cortical activity after in vitro perfusion of cerebral arteries with a voltage-sensitive dye [1]

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Abstract

Cortical neuronal architecture and connectivity can be analyzed with high-resolution optical imaging after staining the in vitro isolated guinea pig brain preparation by circulating the voltage-sensitive dye RH795 via the arterial system. To establish this new technique, electrical field potentials evoked in the piriform and entorhinal cortices by lateral olfactory tract stimulation were correlated to the optical signal. The depth analysis of the optical response was performed by evaluating the contribution of the mono- and poly-synaptic components of the signal generated in different layers after applying a pair-pulse stimulation protocol. The tangential propagation of neuronal activity in olfactory cortices was evaluated by gathering several 4.2 × 4.2 mm images recorded from adjacent cortical areas. The real-time optical imaging technique applied to the isolated guinea pig brain can be successfully utilized to study the integrative properties of cortical neurons ensembles. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Entorhinal cortex; Isolated guinea pig brain; Optical imaging; Piriform cortex; Voltage-sensitive dyes

1. Introduction

The study of neuronal activity propagation have been greatly facilitated since the introduction of real-time optical recordings of membrane potential changes with voltage-sensitive dyes [15]. Tissue staining with fluorescent dyes is obtained in vivo by trans-pial diffusion and in vitro by extracellular superfusion. The penetration of the dye in the tissue after local application on the cortical surface during in vivo experiments is limited by the presence of the pia. Moreover, surface detection of fluorescence generated in the cortex is limited by the opacity of the tissue due to vessels and blood cells. To elude these restrictions we applied the voltage-sensitive dye, RH 795, through the cerebral arterial system in the in vitro isolated guinea pig brain [9,10,12,26,27]. High-resolution optical recordings [19,20] were performed to analyze the distribution and the propagation of the voltage-dependent neuronal signals in this in vitro experimental condition that allows a facilitated access to cortical structures, due to the complete exposure of the brain surface. The optical responses were interpreted and analyzed on the basis of the current understandings of field potentials simultaneously recorded and previously described in the same preparation [5,6]. The contribution of different cortical layers to the generation of the optical signal helped us to determine the depth resolution of the optical recording system and to assess the degree of staining of the isolated brains after arterial perfusion with the voltage-sensitive dye. The results reported here have been communicated in abstract form [21].

2. Materials and methods

Following anesthesia with penthotal-sodium (20 mg/kg) the brain of Hartley guinea pigs (200 g, Charles River) was carefully removed under hypothermic conditions and was transferred to a perfusion chamber, where the basilar artery was cannulated according to the standard procedure [9,12,26,27]. The isolated brain was arterially perfused with a solution containing NaCl 126 mM, KCl 2.3 mM, NaHCO3 26 mM, MgSO4 1.3 mM, CaCl2 2.4 mM, KH2PO4 1.2 mM, glucose 15 mM, HEPES 5 mM and 3%

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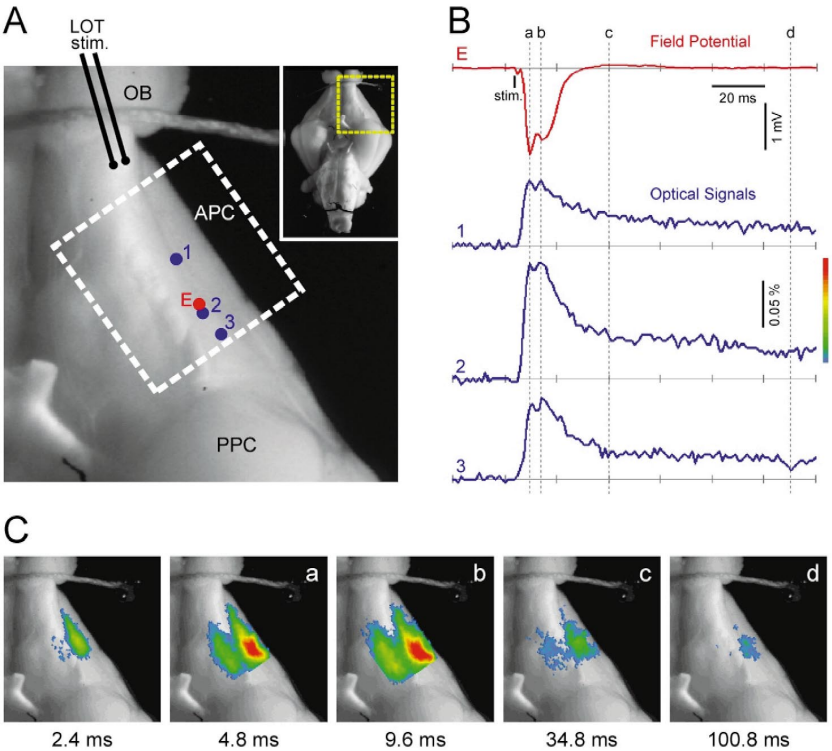

Fig. 1. Correlation between field potentials and optical signals evoked by LOT stimulation in the PC. (A) The area enclosed by the broken square line represents the camera field superimposed to the enlargement of the ventral view of the isolated guinea pig brain, as it appears during the experimental session (inset on the right); OB, olfactory bulb; APC, anterior piriform cortex; PPC, posterior piriform cortex; LOT, lateral olfactory tract. Stimulating electrode (Stim) positioned on the LOT. Field potentials were recorded from E. (B) Correlation between the LOT-evoked field response (E) and the voltage-dependent optical signals recorded from the pixels located in 1, 2 and 3. The vertical lines mark the time point corresponding to the mono-synaptic (a), the di-synaptic (b) peaks and the late components of the responses (c and d). (C) Spatio-temporal pattern of optical changes in the PC represented by pseudo-color image using color code calibration shown in B. Delays from LOT stimulus are indicated below each panel. The distribution pattern corresponding to times a, b, c and d are marked.

dextran 70 000, oxygenated with a 5% CO_2, 95% O_2 gas mixture (pH 7.3). Perfusion rate was 5.5–6 ml/min. Experiments were performed at 27–30°C. The experimental protocol has been reviewed and approved by the Ethical Committee of the Istituto Nazionale Neurologico.

Electrophysiological recordings were performed with 0.9% NaCl filled micropipettes (5–10 MΩ input resistance). Stimulation was delivered through bipolar tungsten electrodes (100 μm between electrodes) positioned on the lateral olfactory tract (LOT). Stimulus intensity and duration were 20–200 μAmp, 100 μs. Simultaneous optical and electrophysiological recordings were performed from the piriform cortex (PC) and the entorhinal cortex (ERC). The voltage sensitive dye RH-795 (1 mg in 20 ml per-

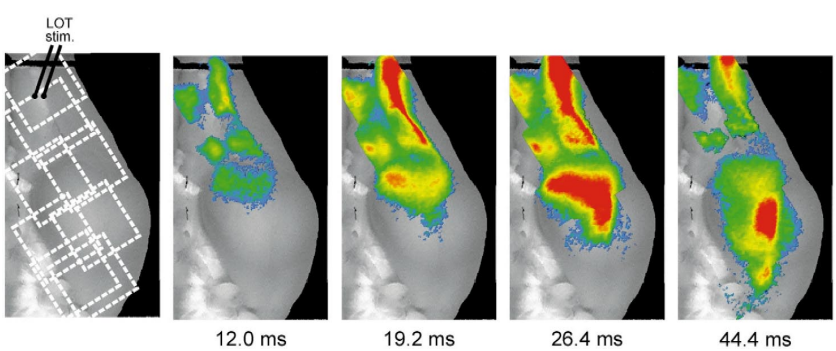

Fig. 2. Neuronal activity propagation in the PC and ERC following LOT stimulation. On the left, the eight different 4.2 x 4.2 mm squares from which optical recording illustrated in the other panels were performed at different sequential times were superimposed to a CCD camera-derived image of the ventral view of the brain. The camera was moved to the next field after the acquisition of the optical signal. The same LOT stimulation intensity was applied to record from an adjacent cortical area. The propagation pattern of optical signal illustrated in the four panels on the right was reconstructed by gathering off-line the individual recordings of each of the 8 sessions. Delays from the LOT stimulus are indicated. The results are also illustrated in Animation 1.

fusate) was arterially perfused for 5–10 min before starting the optical recordings. A single staining of the brain allowed to record large optical signals for 2–3 h. In three experiments the diffusion of the dye was verified by measuring the fluorescence emitted by a 540 μm wavelength light in coronal sections cut from the brain at the end of the recordings. The RH-795 arterial perfusion stained uniformly the cerebral tissue. The optical recording system [19,20] had a time resolution (frame rate) of 0.6 ms. The photodetector was a MOS sensor composed of 128 by 128 elements. To improve the signal-to-noise ratio, we averaged 2 by 2 pixel groups into a single element, thus images are presented as 64 by 64 pixels. The photodetector covered a cortical area of about 4.2 × 4.2 mm when we used two-fold epi-fluorescence microscope; each element thus covered 65 μm^2. At this spatial resolution we could image the structure of the cortical surface as well as record neural activity. The optical apparatus, constructed in our laboratory, was similar in design to the tandem type epi-fluorescent microscope of Ratzlaff and Grinvald [28]. We used a custom-made 25 mm lens (F1.2) as first objective, and a 50 mm lens (F1.85) for second objective, which altogether results in approximately a two-fold magnification. Since the size of the image sensor is slightly less than 1 × 1 cm, this low magnification optical apparatus allowed us to record from 4.2 × 4.2 mm area of the brain. RH-795 has excitation and emission maxima at 530 and 712 nm, respectively. Light from a tungsten-halogen lamp was passed through a heat filter (less than 10% transmittance for 700–1600 nm light) and an interference filter (535 ± 20 nm) before being reflected down onto the brain surface by a dichroic mirror (half-reflectance wave

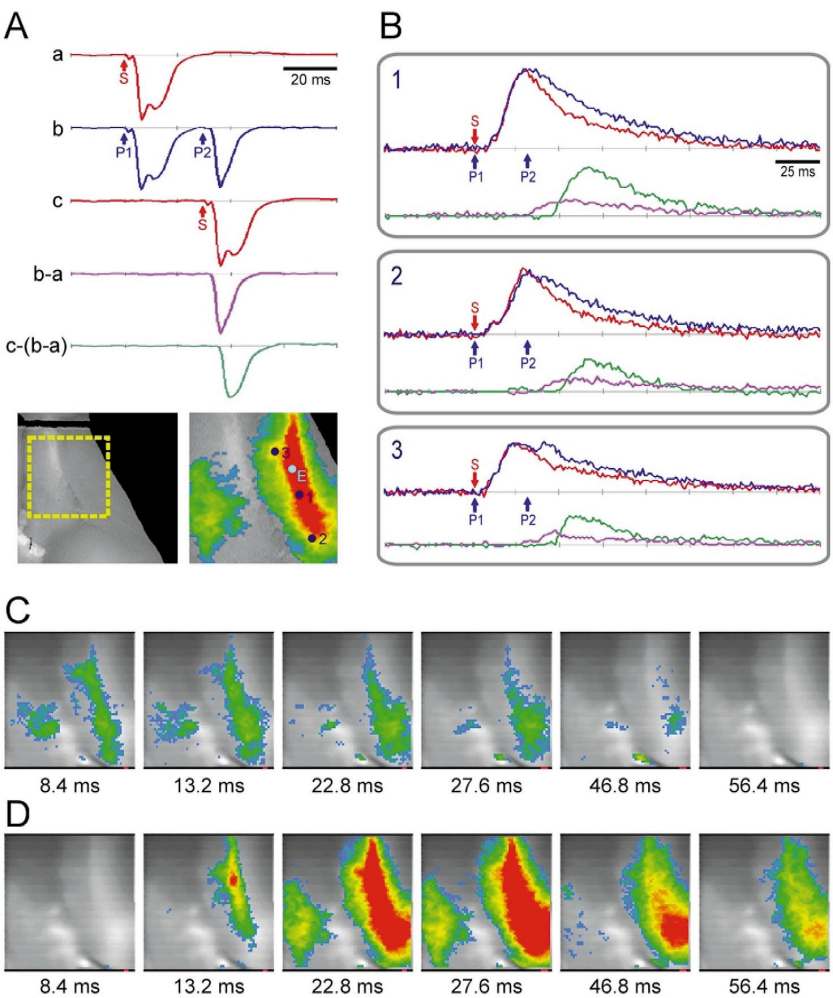

Fig. 3. Isolation of mono- and poli-synaptic LOT-evoked responses in the PC. (A) Paired-pulse paradigm utilized to isolate the synaptic components of the LOT responses. Single stimulation: S. Conditioning and conditioned pairing stimuli: P1 and P2. The color code of the traces is reported in the lower right part of the panel. The lower right panel in A shows the location of the electrophisiological (E) and the optical recording sites illustrated in B. The recorded area is illustrated in the left lower panel in A. (B) For each panel the superimposed upper traces represent the optical signals evoked by single (red) and paired (blue) stimulation. The lower traces show the isolated mono- (purple, b-a) and poli-synaptic (green, c-(b-a)) responses. Two different stimulation intensities were utilized to evoke the electrical field responses in A and the optical responses in B (see text). The spatio-temporal propagation pattern changes of isolated mono-synaptic and poli-synaptic optical signals obtained according to this procedures are illustrated in C and D, respectively. Calibrations are derived from color code bars in Fig. 1(B). The same results are also illustrated in Animation 2.

length of 580 nm). The brain fluorescence was projected to an image sensor through a long-wavelength pass filter (50% transmittance at 600 nm). For each optical image shown, eight to sixteen responses were averaged.

3. Results

Experiments were performed on 21 isolated guinea pig brains. Fig. 1 shows the recording set-up and illustrates the procedure utilized to correlate field potentials and optical signals. The LOT-evoked electrical and optical responses are illustrated in Fig. 1(B) and (C). The field response recorded with the extracellular electrode (E in Fig. 1(A)) in the anterior PC (red trace in Fig. 1(B) was superimposed to the optical signals (blue traces) recorded at different sites within the camera field surrounded by the broken line in Fig. 1(A) (sites 1, 2 and 3). The field potential peaks identified as mono-synaptic (a) and di-synaptic (b) responses [9,18,22] correlated with two optical peak transients. A slowly decaying optical response (time points c and d) followed the mono/di-synaptic peaks.

Electrophysiological studies demonstrated that stimulation of the LOT induces a diffuse activation of the olfactory tubercle, the PC and the ERC [5,7,17,22]. Sequential optical recordings were performed by moving the camera in order to frame different rostro-caudal regions in the PC-ERC after LOT stimulation (Fig. 2 and Animation 1). The 8 fields outlined in the left panel were then merged to describe the cortical propagation of the LOT-evoked activity. As expected, the optical signal appeared first in the anterior PC and the olfactory tubercle, propagated to the posterior PC and traveled in the lateral portion of the ERC, from rostral to caudal.

The mono-synaptic and di-synaptic activity evoked by LOT stimulation is generated by synapses segregated respectively in layers Ia and Ib of the PC, at 0–200 and 200–450 μm depth, respectively. The mono-synaptic component of the field response could be isolated when two LOT stimuli were applied with a 30–40 ms interval [6,23,25]. The poly-synaptic activity was abolished in the conditioned response, as illustrated in Fig. 3(A) (blue trace b). When the paired response was subtracted to the unpaired response, the mono-synaptic potential was isolated (purple trace b-a in Fig. 3A). The poly-synaptic activity was isolated by subtracting the mono-synaptic response (b-a) from an unpaired LOT response evoked with an appropriate delay [Fig. 3A, green trace c-(b-a)]. In Fig. 3B the same subtraction paradigms were utilized to isolate the mono- and the poly-synaptic components in the optical signals recorded at the sites marked in the lower left panel in 3A. The optical recordings do not correspond to the field responses illustrated in A. To isolate an almost purely disynaptic response (and avoid polysynaptic activation and IPSPs components in the optical signal) the stimulus intensity was, indeed, reduced to 1/4–1/5 of the stimulus required to obtain a maximal monosynaptic field response. The purple and green traces in Fig. 3B represent the isolated mono-synaptic and poly-synaptic optical responses, respectively. The spatio-temporal pattern of propagation of the isolated mono-synaptic and poly-synaptic optical signals obtained with this subtraction procedure are illustrated in Fig. 3C and D, respectively, and in Animation 2.

To further evaluate the depth resolution of the optical recording system we evaluated the light emitted at different cortical depths by the tip of a glass pipette filled with Nile Red, a fluorescent substance with similar excitation and emission maxima of RH-795, but stronger fluorescence. Measurements of fluorescence intensity were performed when the tip of the electrode was lowered gradually in the tissue from the cortical surface. As illustrated in Fig. 4, the light intensity normalized to the fluorescence emitted by the tip of electrode positioned at the cortical surface decreased exponentially with the depth of the electrode tip. In the isolated brain, the gradual decrease in optical signal allow to record a 10% fluorescent light intensity even at 1000 μm depth (squares and triangles), whereas in vivo the same value was reached at 700 μm depth (circles), suggesting that light absorption is lower in the isolated brain preparation. Such a difference may be due to the higher transparency in the isolated brain due to the complete washout of blood cells, which are known to absorb fluorescent emitted light. The degree of fluorescence reduction depends on both the defocusing effect and

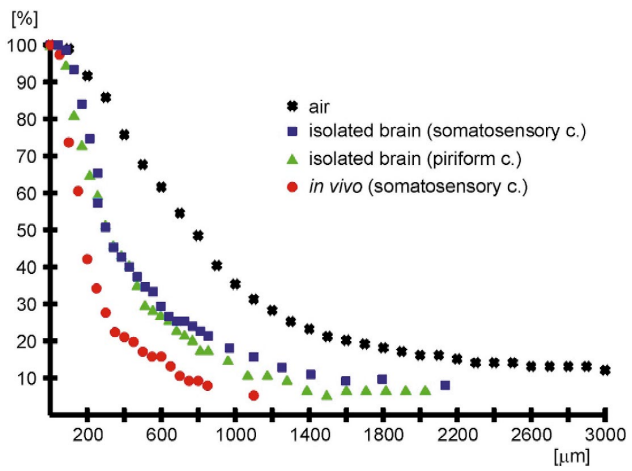

Fig. 4. Normalized optical signal intensity at different depths in the cortical tissue measured at the cortical surface when the tip of a pipette loaded with the fluorescent dye Nile Red was lowered in the cortex (100% corresponds to the surface recording). The focal plane of the microscope was maintained at 0 depth during the measurements. Measurement performed in the PC (triangle) and in the somatosensory cortex (squares) of the isolated brain in vitro, in the somatosensory cortex (circles) of guinea pig brain in vivo and in the air (crosses) are illustrated. The measurements performed in the air (no tissue penetration) reflect the decrease in recorded light due to the defocusing of the electrode tip by moving it from 0 to 3000 μm from the focal plane.

the decrement of the excitation and light emission during the passage of the electrode tip through the cortical tissue. Indeed, when the same test was performed in the air the fluorescence intensity also decreased when the tip of the pipette was defocused (crosses in Fig. 4).

4. Discussion

Voltage-sensitive dyes are usually applied to the nervous system either by trans-pial diffusion in vivo or by extracellular superfusion in sliced tissue maintained in vitro. The penetration of the voltage-sensitive dye in the tissue after local application on the cortical surface during in vivo experiments is limited by the presence of the pia. In addition, surface detection of fluorescence generated in the cortex is limited by the opacity of the tissue due to vessels and blood cells. Recently, optical recordings with a multi-photodiode array system have been applied to the in vitro isolated guinea pig brainstem and cerebellum preparation [8]. As for the in vivo studies, these authors applied a voltage-sensitive dye by external superfusion to study the lateral inhibition and the on-beam inhibition in the cerebellar cortex in response to the parallel fibers stimulation. Our different experimental approach aims at staining the brain diffusely and homogeneously via the vascular system. In such conditions real-time tangential propagation of neuronal activity can be followed with little spatial restrictions.

The depth location of the generators of the optical activity is not immediate and needs to be extrapolated by applying pairing stimulation tests. The results shown here demonstrate that our recording system yields to discriminate neuronal activity generated as deep as 500 μm from the pial cortical surface. Moreover, the experiments illustrated in Fig. 4 demonstrate that the fluorescence emitted by the dye can be recorded as deep as 1000 μm from the pial surface. The resolution of optical responses in the depth axis is difficult to asses and the real value of recordings performed at depth superior to 600 − 700 μm is questionable, because of the transparency of the tissue and the light scatttering/reflection of both emission light and emitted fluorescence by the neural tissue.

Even if optical signals represent voltage changes and field potentials represent current flow, similar, if not coincident peaks and latencies were observed by comparing electrical and optical measurements in the piriform cortex of the isolated guinea pig brain preparation. Optical recordings of cortical activity has been studied with multiple photodiode arrays in the PC in vivo [23–25] and in vitro [29] and in other cortical structures [2,3,16,30,31]. It is commonly believed that voltage-dependent optical signal is generated principally by membrane depolarizations associated to excitatory synaptic activity. Highly synchronous population spikes have been reported to give rise to small amplitude optical responses of very short duration (<5 ms; see [1,4]), that are much faster than the activity observed in our recordings. In addition, when LOT stimulus intensity was increased in our experiments to increase firing synchronization, no fast optical signal imputable to a population spike activity was observed (not shown). Very tiny population spikes, indeed, can be elicited in the PC, suggesting a loose synchronization among neurons in this cortical region [5,22].

A further question regarding the sources of the optical signal concerns the generation of the slow potentials that follow the peak activity correlated to the excitatory synaptic responses. Intracellular recordings in PC neurons demonstrated fast and late depolarizing inhibitory postsynaptic potentials mediated by GABA receptors [14,18] associated to a late sink located in layer II during current source density analysis of field potential profiles [11,17]. Such potentials correlate in time with the slowly decaying late optical signal observed in our experiments (time point c in Fig. 1(B)). Even though a direct pharmacological demonstration has not been carried out in our study, the abolition of the late optical signal in the unconditioned-conditioned subtracted images during LOT pairing confirms that at least part of the signal is poly-synaptic, maybe associated to inhibitory potentials. Another possible source for the late optical signal would be either the glial depolarization consequent to neuronal excitation or the intrinsically generated optical signal [13].

The study confirms that voltage sensitive dyes cross the blood-brain-barrier and can be utilized in the isolated brain preparation to study in the tangential propagation of neuronal activity. It is also shown that a three-dimensional imaging of the cortical activity can be exploited, since deeply and superficially generated optical signals can be identified from surface recordings. These results validate the use of optical recordings in the isolated guinea pig preparation to study brain function in physiological conditions and after the induction of pathological situations, such as epilepsy and ischemia.

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